SoK: A Systematisation of Attacks on Digital Twin Architectures in Embedded and Control Systems

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***Abstract*—The usage of digital twins (DTs) is increasingly popular in the context of Industry 4.0, cyber-physical systems (CPS), Industrial Internet of Things (IIoT), and critical infrastructure. Digital twins enable the creation of virtual representations of physical entities for real-time monitoring, prediction, optimisation, and automated decision making. Nevertheless, the widespread integration of cloud computing, artificial intelligence, machine learning, and IIoT technologies has introduced significant cybersecurity challenges.**

Unlike traditional IT infrastructure, digital twins involve direct interaction between cyber and physical environments, meaning that attacks on virtual systems may directly affect physical operations. Consequently, digital twin systems are vulnerable to cyber-attacks such as false data injection attacks, replay attacks, adversarial machine learning, synchronization attacks, and multi-stage CPS attacks.

This paper presents a Systematisation of Knowledge (SoK) study of the primary attack surfaces and vulnerabilities associated with digital twin systems while analysing potential cyber-attack vectors and their operational impacts.

***Index Terms*—Digital Twins, Cyber-Physical Systems, Industrial Internet of Things, Digital Twin Security, Cybersecurity, Threat Modelling, Adversarial Machine Learning, Industrial Control Systems, Intrusion Detection, Smart Manufacturing, Cloud Security, Edge Computing, False Data Injection, Synchronization Attacks, Resilient Systems, Systematisation of Knowledge (SoK)**

I. Introduction

A. Digital Twin Attacks and Security Analysis

Digital twins have become an integral part of Industry 4.0, cyber-physical systems (CPS), Industrial Internet of Things (IIoT), smart grids, healthcare industry, and industrial automation [1]–[3]. A digital twin is defined as a virtual representation of a real system that continuously exchanges information with its physical counterpart in order to monitor, simulate, predict, and optimise system behaviour [1], [3].

The rapid growth of digital twin technology has been driven by advancements in cloud computing, Internet of Things technologies, edge computing, artificial intelligence, and machine learning [4], [5]. Contemporary digital twin ecosystems represent highly interconnected combinations of sensors, communication networks, predictive analytics, and control systems [5], [6].

Nevertheless, these features also introduce significant cybersecurity risks. The interaction between digital twins and their physical counterparts occurs through synchronization and control mechanisms [1], [7]. Consequently, attackers may target both virtual and physical systems simultaneously, potentially leading to severe disruptions and failures in critical infrastructures [8], [9].

Existing literature demonstrates that digital twin architectures expose a broad attack surface that includes physical devices, communication protocols, cloud infrastructure, machine learning pipelines, and application interfaces [10], [11]. Threats include false data injection attacks, replay attacks, sensor hijacking, adversarial machine learning techniques, malicious models, and synchronization attacks [12], [13]. Patterson et al. demonstrated a white-box adversarial attack against digital twins [14], while Sen et al. replicated realistic multi-stage cyberattacks within a smart-grid digital twin environment [15].

Attacks on embedded devices and industrial sensors represent another major challenge for digital twin systems. Cui et al. demonstrated how firmware modifications could compromise embedded systems [16], while Shepard et al. showed that phasor measurement units are vulnerable to GPS spoofing attacks [17]. Furthermore, synchronization-based attacks may create inconsistencies between physical and virtual systems, potentially resulting in unsafe operational decisions within industrial CPS environments [18], [19].

To mitigate these threats, the research community has proposed intrusion detection systems, anomaly detection frameworks, resilient synchronization mechanisms, and cyber-resilient digital twin architectures [20], [21]. Systems such as ENIGMA [22] and TwinGuard [23] demonstrate how machine learning and explainable security techniques can improve the resilience and security of digital twin environments.

The growing trend towards incorporating AI-based analytics in DT frameworks also introduces challenges relating to the explainability of AI models [19], [22]. The majority of DT frameworks employ advanced machine learning algorithms to conduct activities such as prediction, fault detection, and intelligent optimisation [3], [20]. In many cases, advanced AI models behave like black boxes that hinder understanding about how specific decisions are made. As industrial enterprises increasingly rely on automated decision-making systems based on DT technologies, issues of transparency, explainability, and accountability have emerged as crucial factors [7], [24].

Moreover, the use of digital twin technologies for training, attack simulation, and security analysis is becoming increas-

ingly popular [25], [26]. Security-oriented DT systems can simulate industrial and critical infrastructure environments in virtual settings, enabling organisations to examine attacks in action, assess defensive mechanisms, and investigate cyber-physical weaknesses without affecting real operational systems [15], [21]. This provides major advantages for incident response preparation, cyber range development, and cybersecurity experimentation within industrial environments [25], [27].

While digital twin technology is experiencing rapid growth across multiple domains, several research challenges remain unresolved [28], [29]. Problems associated with secure interoperability, DT orchestration, security evaluation methodologies, and synchronization mechanisms continue to require further security-focused research efforts [7], [26], [30].

Although research in this area has expanded rapidly, the field remains fragmented. Many studies focus only on isolated attack scenarios or domain-specific implementations, while relatively few works attempt to systematically analyse attack propagation, trust assumptions, inter-layer dependencies, and architectural weaknesses [24], [28], [29]. Consequently, there is a strong need for a Systematisation of Knowledge (SoK) study capable of organising and classifying existing research into a unified taxonomy.

The SoK study presented in this paper investigates attacks against digital twin architectures across CPSs, embedded systems, industrial control systems (ICS), and critical infrastructure environments. The work analyses attack surfaces across physical, communication, cloud, AI, and synchronization layers while also identifying key research challenges related to scalability, synchronization security, adversarial robustness, and cyber-physical resilience.

II. Methodology

In this SoK paper, a systematic methodology is adopted to conduct a study on the security issues, possible attack surfaces, and defense mechanisms for Digital Twins in the cyber-physical and industrial field. More precisely, the core aim of the methodology is to conduct a comprehensive study and analysis of the existing body of literature on Digital Twin architecture in order to uncover attack vectors, weaknesses in security, defense approaches, and gaps in research with regards to them.

The first step in the methodology involves carrying out extensive research in various online sources such as IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, arXiv, MDPI, and ResearchGate. The use of key words such as *Digital Twin Security*, *CPS Security*, *IIoT Security*, *Digital Twin Attacks*, *Adversarial Machine Learning*, *Industrial Control Systems*, *Threat Modeling*, and *Synchronization Attacks* was made. Additionally, key surveys on Digital Twin Security threats by Alcaraz and Lopez [1], Digital Twins attacks and defense approaches by Suhail et al. [12], and enabling factors for Digital Twin technology by Mihai et al. [3] were consulted in order to gather information on the background and technical details of DTs.

Post selection for papers through the first filtering phase, further curation for cybersecurity, industrial systems, embedded devices, CPS, AI-based DTs, and attack analysis was carried out. Papers related exclusively to optimization or manufacturing process improvement or non-security oriented applications were considered only if security-oriented architectures were discussed. An additional focus area included papers related to industrial control systems (ICS), smart grids, critical infrastructure, and OT environments, where DTs can be deployed in highly risky scenarios [9], [31].

These selected papers were subsequently categorized into the following three main sections:

1) Surveys/SoK Papers
2) Papers on Attacks and Threat Models
3) Defensive and Resilience Papers

Surveys/SoK papers were primarily utilized for understanding the typical architectures, vulnerabilities in attack surfaces, and recent research trends associated with DTs. Some important citations in this category are Alhamam et al. [28], Jeremiah et al. [2], Empl et al. [24], and Achuthan et al. [29].

Secondly, studies on attacks and threat modeling for DT systems were reviewed to determine the primary weaknesses associated with physical systems, communication protocols, cloud infrastructure, AI pipelines, and synchronization mechanisms. Studies on adversarial attacks and cyber-physical attack methods were crucial for understanding modern threats to DT systems. For example, Patterson et al. conducted a white-box adversarial attack against DT systems [14]. Pooyandeh and Sohn described a timestamp attack against DT-based battery monitoring [18]. Sen et al. replicated a multi-stage cyberattack against smart-grid DT systems [15].

Understanding the nature of attack propagation through DT system architecture required attacks to be categorized based on the target layer of the system. The research methodology revealed that there are five broad DT attack surfaces:

1) Physical and sensor layer
2) Communication and networking layer
3) Cloud and edge infrastructure
4) AI and machine learning pipelines
5) Control and synchronization mechanisms

In each attack category, the study analyzed:

1) The attack technique
2) Attack objectives
3) DT layer attacked
4) Impact of attack on the operation of the DT system
5) Defense mechanisms proposed

For instance, attacks involving false data injection [32], replay attacks [33], GPS spoofing [17], and sensor spoofing attacks [34] were investigated due to their direct impact on the integrity of data physically used by DT systems. Attacks involving firmware exploitation [16], sound-based attacks on gyroscopic sensors [35], and industrial traffic manipulation [36] were also examined in order to study how embedded devices and industrial sensors could be exploited as attack vectors against DT systems.

Special emphasis was placed on artificial intelligence-powered attacks against DT systems since contemporary DT systems make extensive use of machine learning techniques for prediction, maintenance, anomaly detection, and optimization of operations. Evasion, poisoning, and backdoor attacks against machine learning models, which are common AI-powered attacks, were studied through various articles including those written by Anjum et al. [13] and Patterson et al. [14].

Synchronization security issues were explored because synchronization is a crucial component of DT architecture itself. State desynchronization attacks and timing attacks were considered to examine implications arising from discrepancies between the physical world and its virtual representation in relation to unsafe decision making [18], [19]. This topic is especially relevant in industrial CPSs since DT outputs can directly impact the operation of automatic physical systems.

Furthermore, apart from cyber threats and attacks on DTs, defensive and resiliency-related research was also considered. Approaches such as intrusion detection systems, anomaly detection techniques, cyber-resilient DT architecture, and explainable security systems were investigated. Notable examples in this field include ENIGMA [22], TwinGuard [23], digital twin-enabled intrusion detection systems [20], DT-based cyberattack detection mechanisms [37], and cyber-resilient digital twin architecture for critical infrastructure [21].

Another critical aspect of the methodology was assessing how various studies validated their techniques. The majority of DT security papers are based on simulations or artificially generated datasets, while significantly fewer studies have been tested within actual industrial settings or through hardware-in-the-loop testing. Hence, the present SoK also examined the level of realism in the experimental environment, considering the use of:

1) Datasets from industrial facilities
2) CPS testbeds
3) Smart grid simulations
4) Real-life ICS networks
5) Embedded hardware systems

Lastly, all results derived from the selected sources were summarized to create a comprehensive taxonomy of DT attack surfaces, vulnerabilities, and defense strategies. Based on the identified taxonomy, it is possible to determine key research gaps and challenges related to:

1) DT scalability issues
2) Attack propagation between layers
3) Adversarial robustness
4) Time synchronization security
5) Cloud/edge trust assumptions
6) Absence of standardized DT security evaluation methodologies

III. Background and Digital Twin Architecture

Digital twins provide a real-time digital representation of a physical asset, process, or cyber-physical system, allowing the system to be monitored, analysed, and adapted continuously [1]. In embedded and control environments, this capability is important because operational decisions often depend on live telemetry, model-based prediction, and timely actuation. However, the same connectivity that enables digital twins also increases the attack surface: compromised sensors, network channels, models, or control interfaces can cause the twin to represent an inaccurate or unsafe version of the physical system [12], [13]. For this reason, the security analysis in this paper begins with architecture. Without a clear architectural model, it is difficult to explain where trust is placed, how data flows through the system, and how attacks may propagate across layers [8], [11].

A. Generic Digital Twin Architecture

This paper uses a five-layer view of digital twin architecture: the physical layer, communication layer, twin layer, data/AI layer, and control/application layer. This structure provides a practical foundation for later taxonomy work because each layer has different assets, assumptions, and failure modes [2], [3]. Figure 1 summarises this layered model.

The physical layer contains the real-world components that produce or receive cyber-physical signals, including sensors, actuators, embedded devices, programmable logic controllers (PLCs), and edge nodes [16], [34]. Since the digital twin depends on these components to maintain an accurate view of reality, manipulation at this layer can corrupt the entire representation. Examples include sensor spoofing, physical tampering, compromised firmware, and actuator manipulation [17], [35].

The communication layer connects the physical environment to the digital representation. It may include industrial and IoT protocols such as MQTT, OPC-UA, DDS, CAN, gateways, brokers, and cloud/edge links [4], [31]. This layer is security-critical because telemetry, commands, and synchronisation messages pass through it. Attacks such as interception, replay, message injection, and latency manipulation can therefore affect both the state of the twin and the behaviour of the controlled system.

The twin layer maintains the virtual model of the physical process. It includes the synchronisation engine, simulation engine, and state estimation logic [5], [22]. If this layer is corrupted, the digital twin may continue to operate while silently drifting away from the real physical state. This makes twin-layer attacks especially dangerous: the system may appear consistent internally while producing unsafe or misleading predictions.

The data/AI layer supports analytics, machine learning, prediction, and anomaly detection. These functions help the twin transform raw telemetry into operational insight, but they also introduce AI-specific risks such as adversarial inputs, poisoned data, model tampering, and false alerts [19], [23]. Finally, the control/application layer exposes dashboards, orchestration tools, operator interfaces, alerting systems, and command functions [21], [24]. At this layer, access-control failures or unauthorised commands may translate digital compromise into real-world physical consequences.

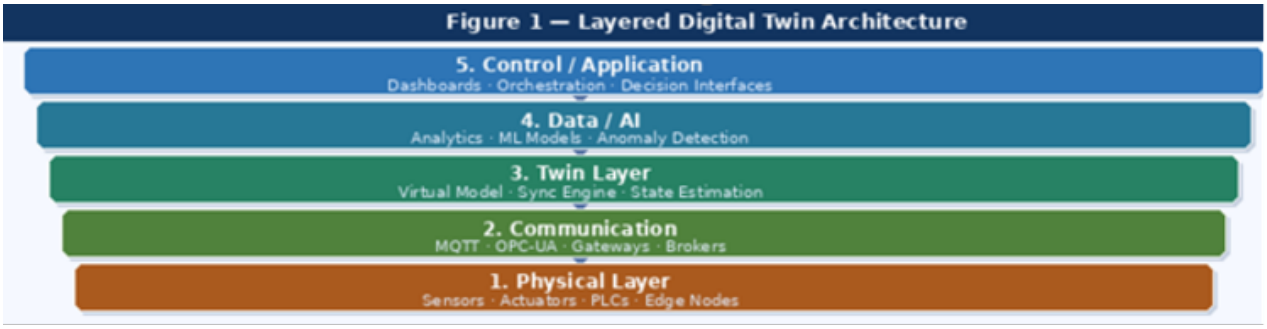


Fig. 1. Five-layer digital twin architecture used as the architectural basis for the attack taxonomy.

TABLE I
FIVE-LAYER DIGITAL TWIN ARCHITECTURE AND PRIMARY SECURITY CONCERNS.

Layer	Key components	Primary security concern
Physical	Sensors, actuators, PLCs, edge nodes	Measurement spoofing and physical/firmware tampering
Communication	MQTT, OPC-UA, gateways, brokers	Replay, interception, injection, and latency manipulation
Twin	Virtual model, synchronisation engine, state estimator	Model drift, state corruption, and desynchronisation
Data/AI	Analytics, ML models, anomaly detection	Adversarial inputs, poisoning, and false alerts
Control/App	Dashboards, orchestration, operator interfaces	Unauthorised commands, access abuse, and audit gaps

TABLE II
REAL-TIME TIMING REQUIREMENTS FOR DIGITAL TWINS IN EMBEDDED AND CONTROL ENVIRONMENTS.

Timing requirement	Definition	Failure consequence
State freshness	The twin reflects the current physical state within an acceptable time lag	Stale state drives incorrect or unsafe decisions
Decision timeliness	Insights reach operators or controllers before the action window closes	Missed interventions and delayed response
Control safety	Commands are issued using valid, current, and authorised state	Unsafe actuation based on outdated or manipulated state

B. Digital Twins in Embedded and Control Systems

Digital twins are increasingly used in embedded and control systems because these environments require continuous situational awareness, rapid diagnosis, and timely decision-making under operational constraints [10], [14]. Unlike purely informational systems, embedded and control systems can affect physical processes directly. As a result, a digital twin attack may not only compromise data confidentiality or availability; it may also influence physical behaviour, safety, and operational continuity [9], [37].

Figure 2 illustrates the synchronisation loop that links the physical system, the digital twin, and the control/application layer. Sensor readings and controller data are transmitted as telemetry to update the twin's model and state estimator. The twin then produces predictions, warnings, or recommendations that support operator decisions or automated control. Feedback and commands may then flow back toward the physical process, completing the loop [19], [33], [36].

This loop creates an important security dependency: the twin is only trustworthy if the data, timing, model state, and command path remain trustworthy. If telemetry is delayed, replayed, modified, or selectively dropped, the twin may remain operational while becoming disconnected from the real system. In that situation, predictions and alerts may appear technically valid but may no longer represent the present physical state [18], [26]. This is why attacks against digital twins should be analysed as propagation paths across architecture, rather than as isolated events.

Real-time behaviour is therefore central to digital twin security. Three timing requirements are particularly important in embedded and control environments: *state freshness*, meaning that the twin reflects the current physical state within an acceptable lag; *decision timeliness*, meaning that insights reach operators before the relevant action window closes; and *control safety*, meaning that commands are issued only from a valid, current, and authorised state [7], [38], [39]. Table II summarises these requirements and the consequences of failure.

Overall, this architectural view establishes the foundation for the threat model and taxonomy developed in the next section. The physical layer defines where real-world signals originate, the communication layer explains how those signals move, the twin and data/AI layers explain how state and insight are produced, and the control/application layer explains how digital outputs can influence action. These dependencies make digital twins powerful, but they also create multi-layer attack paths that are especially relevant to embedded and control-system security.

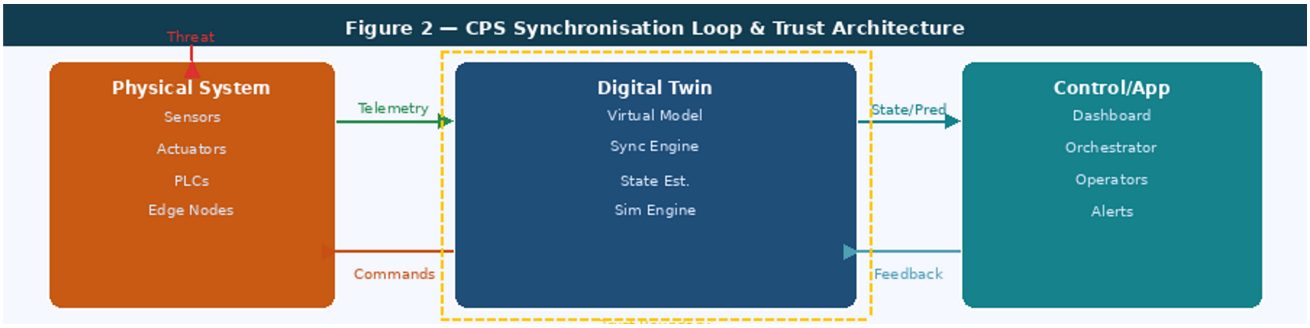


Fig. 2. CPS synchronisation loop showing telemetry, state/prediction output, feedback, commands, and trust boundaries between the physical system, digital twin, and control/application layer.

IV. Layered Taxonomy of Digital Twin Attacks

Digital Twin (DT) systems present an unusually broad attack surface because they combine physical sensing, communication, simulation, machine learning, and cloud-hosted services into a single feedback loop. A threat-led classification must therefore capture both *where* an attack enters the system and *what it affects* in the cyber-physical process. This section first defines the threat model that underpins our analysis, then presents the layered attack taxonomy, and finally synthesises cross-layer, multi-stage adversaries.

A. Assets

We consider five DT assets at risk: (i) live telemetry from sensors and IoT endpoints; (ii) the synchronisation channel between the physical and digital sides; (iii) the twin state and simulation parameters maintained by the engine; (iv) the ML models used for prediction, anomaly detection, and control optimisation; and (v) the control commands, credentials, and APIs that connect operator and machine-to-machine access [1], [12], [13].

B. Attacker Capabilities

We consider four representative capability profiles consistent with the surveyed literature: (a) an *external network attacker* able to observe, inject, or modify traffic on the synchronisation channel [1], [8]; (b) a *malicious insider* with site or operator-level access enabling device tampering or credential abuse [11], [12]; (c) a *supply-chain or firmware attacker* able to plant persistent compromise in sensors or edge gateways [11], [16]; and (d) a *compromised cloud or tenant* able to alter models, configurations, or orchestration state [10], [26], [40].

C. Attacker Knowledge

For ML-targeted attacks we distinguish black-box, grey-box, and white-box adversaries. The DT literature is dominated by white-box and grey-box assumptions, particularly in adversarial-example studies [13], [14], which is itself a limitation we revisit in Section V-D.

D. Attacker Goals

Adversary goals span sabotage of the controlled process, espionage on intellectual property and operational data, manipulation of operator decisions through false twin state, denial of synchronisation, and lateral movement into adjacent OT or IT systems [1], [15], [41].

E. Trust Assumptions and CIA Mapping

The DT operating model implicitly trusts its inputs: sensor readings are assumed real, the synchronisation channel is assumed authentic, the engine is assumed to converge on a faithful state, and the ML pipeline is assumed to be trained on representative data. Each assumption maps to a CIA property whose violation produces a characteristic class of DT-specific harm. *Confidentiality* violations expose live telemetry, trained models and digital blueprints, or identity material, leaking operational state or granting legitimate-looking entry to higher layers [1], [12]. *Integrity* violations can occur at the source (compromised sensor), in transit (modified packets), or at rest (tampered history or models); because each stage feeds the next, a single violation cascades into biased model updates, false state, and incorrect control [22], [32]. *Availability* violations split into endpoint outage, channel partition, and twin-side resource exhaustion; each produces the same operational symptom — the twin ceases to be a reliable mirror of reality — but requires distinct defensive responses [4], [23].

F. Classification Rule

Throughout Section V, we classify attacks by the architectural layer at which the *effect* on the twin manifests, not necessarily the layer of the initial entry point. Attacks that meaningfully traverse multiple layers in a single campaign are treated separately as cross-layer attacks in Section V-F.

G. Taxonomy Axes

Table III structures the surveyed work along four axes: target layer, dominant CIA property violated, attacker capability and knowledge, and DT-specificity (whether the attack is generic to embedded/CPS systems or arises from properties unique to the digital twin paradigm such as synchronisation, bidirectional control, or model-driven decisions). This four-dimensional

view allows us not only to locate each paper but to compare its assumptions against the rest of the corpus.

V. Systematisation of Attacks Across Digital Twin Layers

This section organises attacks on Digital Twin (DT) architectures along the architectural layers they target: Physical, Communication/Network, Twin Engine, AI/ML, and Application/Cloud. Each subsection identifies the assets at risk, the dominant attack patterns reported in the literature, and the CIA properties most frequently violated. A final cross-layer subsection synthesises multi-stage attack chains, in which adversaries traverse several layers to reach the cyber-physical process.

A. Physical Layer Attacks

The physical layer comprises the sensors, IoT endpoints, and edge devices that produce the data stream the twin consumes. An attacker manipulating the physical environment or the hardware doing the sensing can corrupt the twin's view of reality without ever touching its network or compute layers [11], [13]. The simplest case is device tampering, theft, or destruction, which cuts the twin off from part of the system or causes it to receive altered measurements; insiders with site access are the canonical threat actor in industrial settings [11], [31]. More complex are physical-channel sensor spoofing attacks: GPS spoofing of phasor measurement units in power systems [17], acoustic injection on MEMS gyroscopes in drones [35], and active-sensor spoofing against ranging devices [34]. These cause the sensor to report attacker-chosen values that are indistinguishable from genuine readings at the hardware level, defeating any cryptographic protection further up the stack because the data is already wrong before it leaves the device.

The DT-specific harm is that the twin has no independent oracle for the physical state. Every higher-layer protection — encrypted channels, authenticated state updates, anomaly detection on the model — presumes that the sensor readings are genuine. When physical-layer attacks invalidate that assumption, the twin becomes an authoritative source of false information, and its downstream control actions reinforce rather than correct the manipulation [9], [22].

Across the physical-layer literature surveyed, three recurring assumptions limit the strength of current evidence. First, most demonstrations target a single sensor modality (GPS, MEMS, LiDAR) in isolation [17], [34], [35], and DT-side propagation is inferred rather than measured end-to-end. Second, evaluations are dominated by lab testbeds; few studies report results against production-grade twins synchronising under realistic latency budgets.

B. Communication / Network Layer Attacks

The communication channel between physical assets and the digital twin is the most exposed surface of the entire architecture, because every synchronisation update, command, and telemetry packet has to traverse it. The protocol mix typical of a DT deployment (HTTP, MQTT, CoAP, OPC-UA, AMQP, plus wireless bearers such as 4G/5G, ZigBee, Wi-Fi) means at least one hop in the chain is usually weaker than the rest, and that hop is what an attacker targets [1], [4].

Man-in-the-middle (MitM) attacks place an adversary between the controller and the twin, where they can read, modify, or inject packets in either direction. Industrial protocols make this disproportionately easy: Modbus/TCP carries no integrated authentication or integrity protection [31], [36], and IIoT publish–subscribe brokers will often accept traffic from any node that can reach them. For a DT the consequences are bi-directional: modified sensor traffic causes the twin to converge on a wrong model of the plant, while modified control traffic going the other way pushes malicious setpoints down to real actuators [1], [8].

False data injection (FDI) extends MitM by exploiting redundancy in state estimation: Liu et al. [32] showed that an attacker who knows the system topology can craft modifications that pass bad-data detectors in power-grid state estimators. For DTs this matters because the twin itself performs continuous state estimation, and any injection that survives the underlying estimator will be silently absorbed into twin state and propagated to predictive and control services [20], [37].

Replay attacks capture legitimate traffic between the sensor stack and its consumer, then retransmit it later to create sabotage or to trigger actions out of context. The canonical industrial example is Stuxnet, which targeted the Siemens PLCs controlling centrifuges at Natanz; alongside its PLC ladder-logic component, the worm replayed pre-recorded "healthy" sensor readings to the SCADA monitoring interface so that operators observed normal traffic while the centrifuges were driven outside their safe rotational range and physically destroyed [41]. Translated to a Digital Twin environment, the same network-layer pattern has a sharper impact: the twin sits exactly where the SCADA monitor sat, consuming the very telemetry stream that the attacker is replaying, and would synchronise its state model against the falsified readings just as the human operators did at Natanz. Every downstream service — predictive maintenance, anomaly detection, operator dashboards — would then operate on a model confirming a healthy state, masking the sabotage rather than exposing it. The risk is amplified when the twin is bidirectional: an attacker capable of replaying telemetry in one direction can typically replay or modify control packets in the other, allowing manipulation of the simulation state to feed back into real actuator commands without the attacker ever touching the OT network directly. Replay remains effective against any Digital Twin pipeline that does not authenticate the originality of every message via nonces, sequence numbers, or signed timestamps — the canonical defensive requirement for the digital thread [33], [41].

C. Twin Engine Layer Attacks

The Twin Engine layer holds the synchronised state — the live model the digital twin maintains as its representation of the physical system. A desynchronisation attack does not need

TABLE III
LAYERED TAXONOMY OF ATTACKS ON DIGITAL TWIN ARCHITECTURES. CIA: DOMINANT PROPERTY VIOLATED (C/I/A). DT-SPEC.: WHETHER THE ATTACK EXPLOITS DT-SPECIFIC PROPERTIES (SYNCHRONISATION, BIDIRECTIONAL CONTROL, MODEL-DRIVEN DECISIONS) OR IS INHERITED FROM GENERIC EMBEDDED/CPS SETTINGS.

Layer	**Assets at risk**	**Representative attacks**	**Attacker capability / knowledge**	**CIA**	**DT-spec.**	**Key refs**
Physical	Sensors, IoT endpoints, edge devices	Device tampering, theft; sensor spoofing (GPS, LiDAR, acoustic); fault injection	Insider / physical access; external RF or optical adversary	I, A	Partial (inherited from CPS)	[11], [17], [34], [35]
Communication / Network	Sync channel, control packets, MQTT/Modbus/OPC-UA traffic	Man-in-the-middle, false data injection, replay of telemetry / control	External network attacker; on-path adversary	I, A, C	Yes (bidirectional sync)	[31]–[33], [36], [41]
Twin Engine	Synchronised state, simulation parameters	Desynchronisation via selective drop / reorder; timestamp manipulation; state poisoning	On-path or intermediate-node attacker; insider with engine access	I, A	Yes (DT-unique)	[18], [21], [22], [42]
AI / ML	Training data, model weights, inference outputs	Adversarial examples (evasion), data / model poisoning, inference evasion against DT-IDS	White-/grey-box on model; data-pipeline access	I, A	Yes (model-driven control)	[13], [14], [19], [20], [23], [37]
Application / Cloud	APIs, identity, containers, edge–cloud trust boundary	API abuse, credential theft, container/orchestration compromise, malicious model/config rollout	Compromised tenant, stolen credentials, exposed API	C, I, A	Partial (DT amplifies impact)	[1], [7], [10], [12], [26], [40], [43]
Cross-layer	Full DT pipeline	Multi-stage chains: credential theft → telemetry tampering → engine drift → control influence	Persistent, multi-capability adversary	C, I, A	Yes (exploits DT interdependence)	[9], [11], [15], [41]

to break the engine or steal its data; it only has to push the twin's internal state out of synchronisation with reality, after which every downstream service (prediction, monitoring, control feedback, anomaly detection) operates on a model that no longer corresponds to what the physical plant is actually doing [22], [42]. Because the twin is only useful when it mirrors its counterpart within tight temporal bounds, even a small persistent drift defeats the purpose of having a twin at all.

Selective packet drop and reordering achieves desynchronisation without touching timestamps and is one of the possible attack courses on the engine layer. An attacker controlling an intermediate node (a switch, a gateway, or a malicious VM on the data path) can drop a fraction of state updates or deliver them out of order. Because the engine typically interpolates between updates, it fills the gaps with plausible but fictional state transitions, and the twin's state differs from the physical state in a way that often resembles normal sensor noise to an anomaly detector [21], [42].

Timestamp manipulation attacks the synchronisation logic directly. Pooyandeh and Sohn [18] demonstrate a timestamp attack on a DT-based lithium-ion battery monitor for electric vehicles, in which subtly shifted timestamps cause the twin's state-of-charge estimate to drift, illustrating that the twin's notion of "now" is itself an attack surface distinct from packet contents.

The twin engine is the layer where DT-specific risk is highest but evaluation evidence is weakest. Most surveyed studies [21], [22], [42] demonstrate desynchronisation on bespoke testbeds and report qualitative deviation rather than calibrated drift metrics. There is no consensus benchmark for "how desynchronised is too desynchronised", which makes it hard to compare defences.

D. AI / ML Layer Attacks

In the AI/ML layer of the digital twin architecture, attacks focus on the learning methodology that supports anomaly detection, state estimation, simulation calibration, and control optimisation. We define AI/ML layer attacks as those that manipulate the training data, model behaviour, or output used by the twin to represent or control a cyber-physical process. This layer is significant for embedded and CPS deployments, as incorrect estimations can be sent back into safety-critical control choices [1], [13], [14], [22].

Adversarial examples target the inference stage by applying small perturbations to data collection, images, or feature streams so that the Digital Twin model produces an incorrect classification or prediction. Patterson et al. [14] demonstrate a white-box adversarial attack on a DT-enabled automotive CPS, showing how knowledge of the model may be utilised to generate inputs that mislead the twin. This attack is classified as DT-specific when the modified model output affects diagnostics, maintenance choices, or control-loop outcomes.

Poisoning attacks target the training or retraining stage. Many Digital Twins are updated continuously using operational data. An attacker equipped to insert manipulated data into the synchronisation stream can slowly move the model decision boundary. This is especially relevant to industrial DTs and SCADA systems where previous and live data are

reused to enhance prediction models [20], [37]. The learning of misleading relationships by barely correlational models may be further emphasised by constructive Digital Twin work, which occurs when the physical process itself changes under attack or when training samples are manipulated [19].

Inference evasion occurs when an attacker generates operational behaviour that avoids being flagged by a DT-based anomaly detector. This is related to adversarial examples but broader: the attacker may alter process values, network timing, or command sequences so that the DT continues to classify the system as normal [8], [20], [23].

The surveyed AI/ML-layer work shares three recurring assumptions that limit current evidence. First, many studies assume strong attacker knowledge, especially white-box model access [14], which is unrealistic for most deployed DTs. Second, evaluations are typically dataset-limited and do not test adaptive attackers in concrete CPS domains [2], [28]. Third, real-time constraints are under-analysed: an attack that succeeds offline may not be feasible when the twin must process streaming data under tight latency requirements.

E. Application / Cloud Layer Attacks

The application and cloud layer covers the APIs, container platforms, cloud services, and shared infrastructure that host DT functionality. Most modern DT deployments rely on cloud, edge–cloud, or distributed infrastructure for data storage, simulation, analytics, orchestration, visualisation, and remote access, which extends the attack surface well beyond embedded devices and industrial networks [1], [10], [12].

The most recognised danger at this layer is API misuse. Digital Twins provide APIs for querying system state, submitting simulation jobs, obtaining telemetry, or issuing control choices. Weak authentication, lack of authorisation, injection vulnerabilities, and excessive data exposure allow an attacker to read sensitive twin state or manipulate the virtual model. Security surveys consistently identify exposed interfaces and poor access control as recurring environment risks [7], [12], [43]. For embedded and CPS implementations, the impact of a compromised API is extremely critical, because what appears to be a cyber-only vulnerability can affect physical operations once the API is linked to orchestration or control services.

Edge–cloud trust boundaries introduce additional risk. Digital Twins frequently distribute computation between edge devices and cloud back-ends, with edge nodes placed close to sensors and actuators and cloud services providing storage, large-scale analytics, and simulation capacity. A compromised edge gateway may inject false data upstream, while a compromised cloud service may distribute corrupted models or configuration updates downstream. Power-grid and distributed industrial DT experiments reveal that edge–cloud solutions boost flexibility but introduce trust and synchronisation challenges [26], [38], [44].

Credential exploitation involving machine-to-machine credentials, broker passwords, API tokens, certificates, and service accounts allows the attacker to impersonate a legitimate device, service, or operator. This bypasses many network-based defences because the malicious activity appears to originate from a trusted DT component [1], [12]. Credential abuse also supports lateral movement from cloud services into edge gateways.

Container and orchestration risk is increasingly relevant as cloud-hosted DTs adopt microservices and orchestration platforms. Misconfigured role permissions, insecure secrets, exposed dashboards, and vulnerable service meshes allow attackers to move between twin components or tenants [10], [40].

Application/cloud layer attacks differ from physical or sensor attacks because they usually target the management plane of the DT. Their impact still becomes cyber-physical when compromised infrastructure changes models, synchronisation logic, configuration files, or control permissions. The literature is strong at identifying these risks but weaker at experimentally measuring them under production-like deployment conditions, creating a gap between high-level DT security surveys and reproducible cloud/edge attack assessments.

F. Cross-Layer and Multi-Stage Attacks

Realistic adversaries cross several layers in a single campaign. We define multi-stage Digital Twin attacks as attack chains in which compromise of one layer enables progression to another, eventually affecting the integrity, availability, or safety of the cyber-physical process [11], [15], [41].

A typical chain begins with credential theft or network access (Application/Cloud or Communication layer), moves to telemetry manipulation (Communication or Physical layer), corrupts the twin state (Twin Engine layer), and finally influences operator decisions or automated control (AI/ML or Application layer). Sen et al. [15] show how cyber-physical Digital Twins replicate realistic multi-stage smart-grid attacks. Stuxnet, although predating modern DT terminology, remains a canonical reference because it demonstrates how IT compromise, engineering workstation manipulation, and PLC-level payloads combine to produce physical effects [41].

Persistence and lateral movement are the mechanisms that enable layer traversal. Persistence can be implemented at almost any layer: compromised firmware (Physical), modified edge configuration (Communication/Cloud), poisoned datasets or backdoored ML models (AI/ML), or altered simulation parameters (Twin Engine). A DT may continue to reuse corrupted state long after the initial compromise is removed [11], [16]. Lateral movement is facilitated by the trust-by-design pattern: a compromised sensor is trusted by the engine, the engine by the analytics layer, and the analytics layer by the control or application layer [9], [42].

Cascading failures are the cyber-physical outcome of layer traversal. False telemetry accepted by the twin is passed into analytics and then used for control or maintenance decisions, producing wider physical or organisational effects. CPS attack surveys and DT smart-grid studies both show that cascading effects are central to cyber-physical risk, particularly in power, industrial, and critical infrastructure domains [8], [15], [39].

Multi-stage analysis exposes a limitation in much of the DT security literature: many papers evaluate one attack type against one layer, while realistic adversaries combine layers. A SoK taxonomy should therefore classify attacks not only by entry layer but also by propagation path. In our taxonomy, multi-stage attacks represent the highest-sophistication category because they exploit DT interdependence rather than a single isolated vulnerability.

VI. Comparative Analysis

Section IV located individual attacks within the five-layer architecture; this section compares them across that architecture. We first contrast layers along realism, scalability, stealth, and DT-specificity (Section VI-A, Table IV); we then examine how the surveyed literature evaluates these attacks (Section VI-B); and finally we discuss the cross-layer propagation patterns that distinguish DT attack chains from conventional CPS or IT incidents (Section VI-C).

A. Comparison Across Digital Twin Attack Layers

Our layered taxonomy (Table III) shows that attacks on Digital Twins differ not only by entry point but also by realism, scalability, stealth, and DT-specificity. Table IV summarises this comparison and the remainder of this section discusses each layer in turn.

Physical-layer attacks such as sensor spoofing, GPS spoofing, and firmware compromise are among the most realistic because they target the trusted source of telemetry Once corrupted measurements enter the digital thread, neither authenticated communication nor cryptographic protection further up the stack can restore correctness [1], [12]. However, these attacks typically require physical proximity, specialist equipment, or access to the operational environment, which limits their scalability [11].

Communication-layer attacks — including replay, man-in-the-middle manipulation, false data injection, and latency manipulation — exploit the synchronisation channel between the physical asset and its twin [32]. They are generally more scalable than physical attacks because they can be automated and repeated, and their impact is amplified in DT systems: stale, delayed, or replayed telemetry can leave the virtual model internally consistent while becoming externally inaccurate. Their feasibility nevertheless depends heavily on protocol security, network segmentation, and attacker position [4], [31].

Twin-engine attacks form the most DT-specific category. Unlike attacks inherited from generic CPS settings, desynchronisation, timestamp manipulation, and state poisoning directly target the synchronised virtual state and state-estimation logic that distinguish a Digital Twin from a conventional monitoring system [18], [42]. Their main danger is stealth: the twin may continue operating normally while gradually drifting away from the real physical process [19]. Empirical support for this category is, however, weaker than for physical or network attacks, since many studies evaluate desynchronisation on simplified testbeds rather than production-scale DT deployments [22], [42].

AI/ML-layer attacks exploit the growing use of learning models for prediction, anomaly detection, optimisation, and automated decision support [13], [14], [20]. Adversarial examples, poisoning, backdoors, and inference evasion can be highly stealthy because they may degrade decisions without visibly disrupting the underlying system. Their evaluation, however, is often less realistic than that of lower-layer attacks: many studies rely on offline datasets, white-box assumptions, or static benchmarks rather than streaming CPS environments [8], [14]. This creates uncertainty about how well such attacks transfer to deployed DT systems with latency, safety, and resource constraints [2], [28].

Application and cloud-layer attacks provide the greatest potential scale. Compromising APIs, credentials, orchestration services, or edge–cloud infrastructure may allow an attacker to affect multiple twins simultaneously [10], [43]. These attacks are not always unique to DTs, but DT architectures amplify their consequences because cloud-side changes to configurations, models, or access permissions can propagate into cyber-physical decisions [38], [44]. As such, application/cloud compromise represents a bridge between conventional IT security failures and physical operational impact [7], [24].

Overall, the comparison suggests a clear trade-off between realism and scalability. Physical attacks are highly realistic but poorly scalable; cloud and network attacks are scalable but depend on access assumptions; twin-engine and AI attacks are the most DT-specific but currently the least mature in experimental validation. This distinction matters because it shows that DT security cannot be assessed using a single-layer threat model. The most serious risks emerge when attacks move *across* layers — for example, from credential compromise to telemetry manipulation, then to twin-state drift and unsafe control decisions [9], [15], [41].

B. Evaluation Realism and Experimental Limitations

A recurring trend across the surveyed literature is the inconsistency in experimental realism used to evaluate DT attacks. While some studies employ hardware testbeds, industrial devices, CPS simulations, or hardware-in-the-loop environments [17], [35], a substantial portion relies primarily on synthetic datasets, simplified architectures, or theoretical attack models [2], [8], [14]. As a result, the practical feasibility of many proposed attacks remains difficult to assess under realistic deployment conditions.

Physical-layer evaluations are typically the most realistic because they frequently involve direct interaction with sensors, embedded devices, or industrial equipment. GPS-spoofing experiments against phasor measurement units [17] and acoustic attacks on MEMS gyroscopes [35] were validated on real hardware, accounting for environmental noise, timing behaviour, and physical constraints. Similarly, active-sensor spoofing studies use challenge–response protocols on real ranging devices [34]. These works provide stronger evidence of operational feasibility than purely simulated counterparts.

In contrast, AI/ML-focused attacks are generally evaluated under significantly more controlled conditions. Adversarial-

TABLE IV

CROSS-LAYER COMPARISON OF DT ATTACKS ALONG THE FOUR DIMENSIONS DISCUSSED IN SECTION VI-A. REALISM = EASE OF VALIDATING THE ATTACK ON REPRESENTATIVE HARDWARE/SOFTWARE; SCALABILITY = EASE OF AUTOMATING AND REPLICATING ACROSS DEPLOYMENTS; STEALTH = LIKELIHOOD OF REMAINING UNDETECTED BY TYPICAL DT MONITORING; DT-SPECIFICITY = DEGREE TO WHICH THE ATTACK EXPLOITS DT-UNIQUE PROPERTIES (SYNCHRONISATION, BIDIRECTIONAL CONTROL, MODEL-DRIVEN DECISIONS); EVIDENCE MATURITY = PREVALENCE OF END-TO-END EMPIRICAL VALIDATION IN THE SURVEYED LITERATURE.

Layer	Realism	Scalability	Stealth	DT-specificity	Evidence maturity
Physical	High	Low	Medium	Partial (inherited from CPS)	High
Communication / Network	Medium–High	High	Medium	High (bidirectional sync)	Medium–High
Twin Engine	Medium	Medium	High	Highest (DT-unique)	Low
AI / ML	Low–Medium	Medium	High	High (model-driven control)	Low–Medium
Application / Cloud	Medium	Highest	Medium	Partial (DT amplifies impact)	Medium
Cross-layer chains	Medium	Medium	Highest	Highest (exploits DT interdependence)	Low

machine-learning studies commonly assume white-box access to models, unrestricted visibility into the DT pipeline, or access to static offline datasets [13], [14]. Such assumptions simplify experimentation but reduce deployment realism, because production DT systems typically operate under streaming telemetry, resource constraints, and limited attacker visibility [19], [20].

Another recurring limitation is the absence of large-scale distributed DT environments in experimental evaluations. Most surveyed works evaluate attacks against isolated DT components or small laboratory deployments rather than industrial-scale ecosystems consisting of cloud infrastructure, edge nodes, embedded devices, and multiple synchronised twins [38]. Consequently, scalability challenges, network congestion, synchronisation drift, and cascading operational failures are insufficiently explored.

Real-time behaviour also remains under-analysed across the literature. Many DT deployments operate under strict timing requirements where delayed telemetry, stale state estimation, or late control commands may produce unsafe operational outcomes [39]. Yet numerous evaluations measure attack success only in terms of classification accuracy, detection rates, or state deviation, while overlooking latency, timing drift, and operational continuity [18], [42].

Attacker assumptions are also frequently overstated. Several studies assume powerful adversaries with unrestricted access to communication channels, cloud infrastructure, or internal models, without clearly explaining how such access is achieved [13]. While these assumptions are useful for stress-testing DT architectures, they may overestimate attacker capabilities in practical industrial environments where segmentation, authentication, and safety controls are present [1], [31].

Taken together, the literature exhibits a clear imbalance between conceptual attack design and deployment-oriented validation. Existing studies successfully identify important DT attack surfaces and vulnerabilities, but comparatively few works evaluate attacks under realistic cyber-physical conditions involving heterogeneous devices, operational constraints, and long-term synchronisation behaviour [24], [45].

C. Cross-Layer Propagation Trends

One of the most important findings across the surveyed literature is that DT attacks rarely remain isolated within a single architectural layer. Unlike conventional IT systems, DT architectures tightly couple physical devices, communication infrastructures, synchronisation engines, AI-driven analytics, and cloud orchestration services into a continuous cyber-physical feedback loop [3]. Consequently, compromise at one layer frequently propagates into several others.

The communication layer often acts as the primary propagation bridge. Manipulated telemetry, replayed packets, or delayed synchronisation messages can first corrupt the twin engine, which then influences AI-driven prediction systems and operational decision-making [32], [33]. In many cases, the final physical impact is not caused directly by the initial compromise but emerges through gradual propagation across interconnected DT services [9], [15].

Several recurring propagation patterns appear in the literature. A common sequence begins with credential compromise or communication-layer access, followed by telemetry manipulation, state desynchronisation, and eventually unsafe control actions or incorrect operational decisions [11], [15]. Similarly, false data injection at the physical layer may later become poisoned training data for machine-learning models, allowing attacks to persist within DT analytics pipelines even after the original compromise is removed [13].

Cross-layer dependency significantly increases attack complexity compared with traditional CPS environments. A vulnerability that appears low-impact in isolation may become highly dangerous once propagated through synchronisation and automated decision systems. For example, small timing deviations at the communication layer may eventually produce incorrect predictions, delayed anomaly detection, or unsafe actuator behaviour at higher layers [18], [42].

Despite the importance of propagation behaviour, many studies still evaluate attacks within a single subsystem. Experimental designs frequently isolate sensor compromise, AI manipulation, or cloud exploitation without analysing how these attacks influence the broader DT ecosystem over time [14], [44]. As a result, the literature still lacks comprehensive end-to-end evaluations capable of modelling realistic attack chains

across embedded devices, communication protocols, synchronisation services, AI pipelines, and cloud infrastructures [24], [26].

The increasing integration of AI, edge computing, and cloud orchestration into DT environments further amplifies propagation risk. Modern DT systems are highly interconnected, meaning that compromise of one trusted component may cascade across multiple dependent services [7]. The surveyed literature therefore suggests that future DT security research must move beyond isolated attack evaluation toward systemic analysis of multi-layer attack propagation and cyber-physical cascading effects [29], [30], [45].

These observations collectively indicate that Digital Twin security research remains fragmented across architectural layers, evaluation methodologies, and deployment assumptions, motivating the research gaps and future directions discussed in the following section.

VII. Research Gaps and Future Directions

The taxonomy in Section IV and the comparative analysis in Section VI expose several recurring weaknesses in the current Digital Twin security literature. This section synthesises those weaknesses into five open research directions that, in our view, will shape DT security research over the next few years.

A. Lack of DT-Specific Threat Models

Most threat models used in the surveyed literature are inherited from generic CPS or ICS settings rather than designed around DT-unique properties such as synchronisation, bidirectional control, and model-driven decisions. Existing taxonomies inventory the relevant attack surfaces but remain largely descriptive: they list assets and vulnerabilities without expressing how compromise of one component formally enables compromise of another [1], [12], [13].

The synchronisation channel is treated only informally. Few studies model how timing drift, packet reordering, or selective drop interact with the twin's state estimator, even though these primitives are central to DT-specific attacks. A second consequence is that the literature lacks formal attack graphs capable of representing DT interdependence end-to-end — from credential compromise through telemetry tampering, engine drift, and unsafe control action [15], [41]. Future work should therefore develop executable, layered threat models that treat synchronisation semantics, fidelity levels, and digital-thread dependencies as first-class modelling primitives.

B. Limited Realistic Evaluation

As discussed in Section VI-B, much of the DT security literature evaluates attacks in simulation or on small bespoke testbeds rather than under realistic deployment conditions. Hardware-in-the-loop evaluations are concentrated in the physical layer and remain rare for twin-engine, AI/ML, or cloud-side attacks [42].

Distributed DT deployments are similarly underrepresented. Most experiments target single-site, single-twin configurations, leaving multi-site synchronisation, edge–cloud co-ordination, and cascading failure largely untested [38]. Real-time analysis is also weak: many evaluations report classification accuracy or state deviation without measuring latency, timing drift, or operational continuity, even though these are precisely the metrics that determine whether an attack causes unsafe physical behaviour. Closing this gap will require open hardware-in-the-loop testbeds, shared evaluation harnesses for distributed twins, and explicit timing benchmarks alongside the usual detection metrics.

C. AI/ML Robustness Challenges

AI/ML components are now embedded in most DT pipelines for prediction, anomaly detection, and control optimisation, yet the adversarial literature remains dominated by white-box assumptions and offline datasets [14]. These assumptions simplify experimentation but rarely reflect the constraints of deployed DTs, which operate under streaming telemetry, bounded compute, and limited attacker visibility.

Adversarial ML defences designed for static image or text models also transfer poorly to DT pipelines that mix sensor streams, model output, and control actions inside a closed loop. A further problem is explainability: in operational technology environments, an alert from a DT-side detector is only actionable if operators can understand and audit the reasoning behind it [22]. Future work should therefore study adaptive grey- and black-box adversaries against streaming DT pipelines, and pair adversarial robustness with explanations calibrated for OT operators rather than ML researchers.

D. Dataset and Benchmark Deficiency

A persistent obstacle to comparable DT security research is the absence of DT-specific datasets and benchmarks. Most DT-side intrusion-detection work is still validated on legacy ICS datasets such as those of Morris and Gao [36], which predate modern DT architectures and capture neither synchronisation traffic nor twin-state evolution. As a result, detection results reported across papers are difficult to compare and may overstate effectiveness against attacks the underlying datasets were never designed to represent.

There is also no shared evaluation methodology for DT security. Studies report a heterogeneous mix of accuracy, drift, deviation, or latency metrics on different testbeds, with no agreed thresholds for "how desynchronised is too desynchronised". Future work should produce open DT-specific datasets that include synchronisation traffic, twin-state traces, and labelled cross-layer attack chains, alongside benchmark tasks that cover detection, robustness, latency, and lifecycle properties together.

E. Distributed DT Governance and Trust

Modern DT deployments are increasingly distributed across multiple owners, edge sites, and cloud tenants, which introduces governance and trust problems that current security research has only begun to address [26]. In multi-owner settings, no single party has full visibility into the twin's data flows, models, or control surfaces, and lateral movement

between domains becomes difficult to model with single-tenant threat frameworks.

Edge–cloud trust boundaries amplify the problem. A compromised edge gateway may inject false data upstream, while a compromised cloud service may distribute corrupted models or configuration updates downstream. Container and orchestration platforms add further attack surface through misconfigured role permissions, exposed dashboards, and insecure secrets [10]. Beyond the technical layer, there remain open questions about liability, federation, and assurance: when a federated DT influences an unsafe control action, it is often unclear which owner is accountable, which audit trail is authoritative, and how trust should be revoked across the federation [7]. Addressing these issues will require federated trust frameworks, cross-domain audit mechanisms, and standardisation efforts that treat governance as part of DT security rather than as an organisational afterthought.

VIII. Conclusion

This paper has systematised the literature on attacks against Digital Twin (DT) architectures in embedded and control systems by organising the surveyed work around a five-layer architectural model, physical, communication, twin engine, AI/ML, and application/cloud, and a four-axis classification capturing target layer, dominant CIA violation, attacker capability and knowledge, and DT-specificity. This structure makes visible difference the broader literature often leaves hidden. A subset of threats is inherited from generic CPS settings, while another desynchronisation, model-driven decision corruption, and bidirectional control manipulation arises from properties unique to the DT pipeline itself. The cross-layer treatment in Section V further shows that realistic compromise rarely remains confined to a single layer. The trust-by-design relationships between sensors, the synchronisation engine, the analytics pipeline, and the control plane form propagation paths along which a single foothold can become into cyber-physical impact. The preceding comparative analysis and identified research gaps situate these findings within the existing body of DT security work and clarify where the most important open problems lie, particularly around synchronisation security, calibrated drift metrics, and reproducible evaluation in production grade settings rather than custom tests. The contribution of this SoK is therefore not the discovery of new attack primitives but a structured account of how known primitives combine in the DT setting, an account intended to support more precise threat modelling, more comparable defensive evaluations, and a clearer research agenda for resilient digital twin architectures in embedded and control environments.

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APPENDIX

A. Collaboration Reflection

The group coordinated the SoK paper through weekly meetings, shared document editing, and task-based division of work. The project was organised around the main structure of the paper: background and architecture, methodology, layered attack taxonomy, communication/network attacks, twin-engine and synchronisation attacks, AI/ML attacks, application/cloud attacks, and cross-layer analysis. This allowed each member to contribute to a focused part of the literature review while keeping the overall paper aligned with the shared taxonomy.

The Division were made 2x2 where one team covers the architecture and background of Digital Twin and another team covers Threats and Attack Surface. Meeting were contrebuted by using Teams, Whatsapp group,

The key lesson from the collaboration was that a SoK paper requires more than dividing papers between members. The strongest sections came from comparing attack assumptions, affected layers, assets at risk, and evaluation limitations across the literature. Regular meetings helped the group move from individual paper summaries toward a more coherent systematisation of Digital Twin attack surfaces.

B. Group Meeting Log

Date	**Duration**	**Participants**	**Discussion / Outcome**	**Task Assignment / Progress**
24 Apr 2026	20 min	Rahim, Ramiz, Sujeet, Praveen	Initial planning meeting. Agreed the topic focus on attacks against Digital Twin architectures in embedded and control systems. Discussed the need for a layered taxonomy and identified initial search areas.	Allocated literature search tasks across Digital Twin security surveys, CPS attacks, communication-layer threats, AI/ML attacks, and cyber-resilient Digital Twin papers.
1 May 2026	20 min	Rahim, Ramiz, Sujeet, Praveen	Reviewed collected papers and refined the paper structure. Agreed to organise the SoK around five layers: physical, communication/network, twin engine, AI/ML, and application/cloud.	Members began drafting assigned sections. Taxonomy table and architecture figures were planned. References were collected and grouped by attack layer.
8 May 2026	35 min	Rahim, Ramiz, Sujeet, Praveen	Progress review of the draft. Discussed overlap between communication-layer replay attacks, synchronisation attacks, and twin-engine desynchronisation. Reviewed whether each section clearly explained assets, attacker capabilities, impact, and limitations.	Updated draft sections, added cross-layer attack discussion, improved methodology section, and checked that figures/tables supported the taxonomy.
14 May 2026	35 min	Rahim, Ramiz, Sujeet, Praveen	Final review meeting before submission. Clarified remaining content issues, checked consistency of terminology, reviewed citations, figures, tables, formatting, and submission requirements.	Final edits were assigned: proofreading, reference formatting, appendix preparation, checking IEEE template compliance, and preparing the final PDF for submission.

TABLE V
SUMMARY OF GROUP MEETINGS, DISCUSSION POINTS, AND TASK PROGRESS.